TSM 098 | INDEPENDENT PRACTICE

Simplifying fractions

While TSM 099 studies Topic 6. Review: Topic 4.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: simplifying fractions

1. Write in lowest terms.

$$\frac{18}{24}$$

2. Write in lowest terms.

$$\frac{21}{35}$$

3. Write in lowest terms.

$$\frac{28}{42}$$

4. Write in lowest terms.

$$\frac{45}{60}$$

5. Write in lowest terms.

$$\frac{48}{72}$$

6. Write in lowest terms.

$$\frac{54}{90}$$

7. Write in lowest terms.

$$\frac{64}{80}$$

8. Write in lowest terms.

$$\frac{84}{126}$$

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Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: signs and further reduction

9. Write in lowest terms.

$$\frac{-24}{36}$$

10. Write in lowest terms.

$$\frac{30}{-45}$$

11. Write in lowest terms.

$$\frac{-42}{-56}$$

12. Write in lowest terms.

$$\frac{0}{19}$$

13. Write in lowest terms.

$$\frac{96}{144}$$

14. Write in lowest terms.

$$\frac{150}{210}$$

15. Write in lowest terms.

$$\frac{81}{54}$$

16. Write in lowest terms.

$$\frac{128}{160}$$

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Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: equivalent fractions

17. Find the missing numerator.

$$\frac{2}{3} = \frac{?}{12}$$

18. Find the missing numerator.

$$\frac{3}{5} = \frac{?}{30}$$

19. Find the missing numerator.

$$\frac{-4}{7} = \frac{?}{21}$$

20. Find the missing numerator.

$$\frac{5}{8} = \frac{?}{56}$$

21. Find the missing numerator.

$$\frac{7}{9} = \frac{?}{36}$$

22. Find the missing numerator.

$$\frac{0}{5} = \frac{?}{30}$$

23. Find the missing numerator.

$$\frac{11}{6} = \frac{?}{30}$$

24. Find the missing numerator.

$$\frac{-3}{10} = \frac{?}{80}$$

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Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: compare and explain

25. Are these fractions equivalent? Justify.

$\frac{6}{8}$ and $\frac{9}{12}$

26. Are these fractions equivalent? Justify.

$\frac{4}{9}$ and $\frac{8}{15}$

27. Are these fractions equivalent? Justify.

$\frac{-3}{5}$ and $\frac{9}{-15}$

28. Are these fractions equivalent? Justify.

$\frac{12}{18}$ and $\frac{3}{4}$

29. Write in lowest terms.

$\frac{252}{420}$

30. Write in lowest terms.

$\frac{-165}{225}$

31. Explain why dividing only the numerator by 3 changes the value.

$\frac{9}{12}$

32. Can multiplying both numerator and denominator by zero produce an equivalent fraction? Explain.

TSM 098 | ANSWER KEY

Simplifying fractions

While TSM 099 studies Topic 6. Review: Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: simplifying fractions

1. Write in lowest terms.

$$\frac{18}{24} = \frac{3}{4}$$

2. Write in lowest terms.

$$\frac{21}{35} = \frac{3}{5}$$

3. Write in lowest terms.

$$\frac{28}{42} = \frac{2}{3}$$

4. Write in lowest terms.

$$\frac{45}{60} = \frac{3}{4}$$

5. Write in lowest terms.

$$\frac{48}{72} = \frac{2}{3}$$

6. Write in lowest terms.

$$\frac{54}{90} = \frac{3}{5}$$

7. Write in lowest terms.

$$\frac{64}{80} = \frac{4}{5}$$

8. Write in lowest terms.

$$\frac{84}{126} = \frac{2}{3}$$

TSM 098 | ANSWER KEY

Simplifying fractions

While TSM 099 studies Topic 6. Review: Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: signs and further reduction

9. Write in lowest terms.

$$\frac{-24}{36} = -\frac{2}{3}$$

10. Write in lowest terms.

$$\frac{30}{-45} = -\frac{2}{3}$$

11. Write in lowest terms.

$$\frac{-42}{-56} = \frac{3}{4}$$

12. Write in lowest terms.

$$\frac{0}{19} = 0$$

13. Write in lowest terms.

$$\frac{96}{144} = \frac{2}{3}$$

14. Write in lowest terms.

$$\frac{150}{210} = \frac{5}{7}$$

15. Write in lowest terms.

$$\frac{81}{54} = \frac{3}{2}$$

16. Write in lowest terms.

$$\frac{128}{160} = \frac{4}{5}$$

TSM 098 | ANSWER KEY

Simplifying fractions

While TSM 099 studies Topic 6. Review: Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: equivalent fractions

17. Find the missing numerator.

$\frac{?}{8}$

Multiply numerator and denominator by 4.

18. Find the missing numerator.

$\frac{?}{18}$

Multiply numerator and denominator by 6.

19. Find the missing numerator.

$\frac{?}{-12}$

Multiply numerator and denominator by 3.

20. Find the missing numerator.

$\frac{?}{35}$

Multiply numerator and denominator by 7.

21. Find the missing numerator.

$\frac{?}{28}$

Multiply numerator and denominator by 4.

22. Find the missing numerator.

$\frac{?}{0}$

Multiply numerator and denominator by 6.

23. Find the missing numerator.

$\frac{?}{55}$

Multiply numerator and denominator by 5.

24. Find the missing numerator.

$\frac{?}{-24}$

Multiply numerator and denominator by 8.

TSM 098 | ANSWER KEY

Simplifying fractions

While TSM 099 studies Topic 6. Review: Topic 4.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: compare and explain

25. Are these fractions equivalent? Justify.

$\frac{3}{4}$ and $\frac{3}{4}$

Yes; the reduced values match.

26. Are these fractions equivalent? Justify.

$\frac{4}{9}$ and $\frac{8}{15}$

No; the reduced values differ.

27. Are these fractions equivalent? Justify.

$-\frac{3}{5}$ and $-\frac{3}{5}$

Yes; the reduced values match.

28. Are these fractions equivalent? Justify.

$\frac{2}{3}$ and $\frac{3}{4}$

No; the reduced values differ.

29. Write in lowest terms.

$\frac{252}{420} = \frac{3}{5}$

30. Write in lowest terms.

$\frac{-165}{225} = -\frac{11}{15}$

31. Explain why dividing only the numerator by 3 changes the value.

$\frac{9}{12} \neq \frac{3}{12}$

The original is 3/4; the new fraction is 1/4. Divide both parts by the same nonzero factor to preserve value.

32. Can multiplying both numerator and denominator by zero produce an equivalent fraction? Explain.

No. It produces 0/0, which is undefined.